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Physical and Biological Oceanographic Responses to Hurricane Ian on the Florida Shelf

I. ABSTRACT

In late September 2022, Hurricane Ian intensified into a Category 4 storm over the southeastern Gulf of Mexico before making landfall near Fort Myers, Florida on September 28. Beyond its devastating impact to local communities, Ian's path over the West Florida Shelf triggered a series of air-sea interactions, resulting in a distinct "cold wake" and subsequent biological responses along the hurricane's path. This study utilizes satellite-derived Sea Surface Temperature (SST) and Chlorophyll *a* concentrations to determine how this high-intensity event directly impacted the physical and biological characteristics of the regional marine ecosystem. Analysis reveals that the Gulf of Mexico experienced a thermal anomaly, with SST dropping by 3.6°C along the storm's path. While the initial physical forcing occurred during the storm's passage, the 8-day composite data indicates a temporal alignment between the largest cold wake and the peak phytoplankton bloom. At the site of maximum intensity of the storm on the West Shelf, Chlorophyll *a* concentrations increased by 555%. In contrast, the Atlantic coast showed a much more muted response, likely due to the storm's weakening over land and differences in regional bathymetry. These findings suggest that while vertical mixing and cooling are immediate, the resulting biological response remains measurable and significant throughout the week following the event. As climate change increases the frequency of high intensity storms, understanding these interactions is essential for monitoring the long-term ecological health and carbon cycling of our oceans.

APPENDIX

Figure 1: Mapped Ian Hurricane Track (SEADAS)

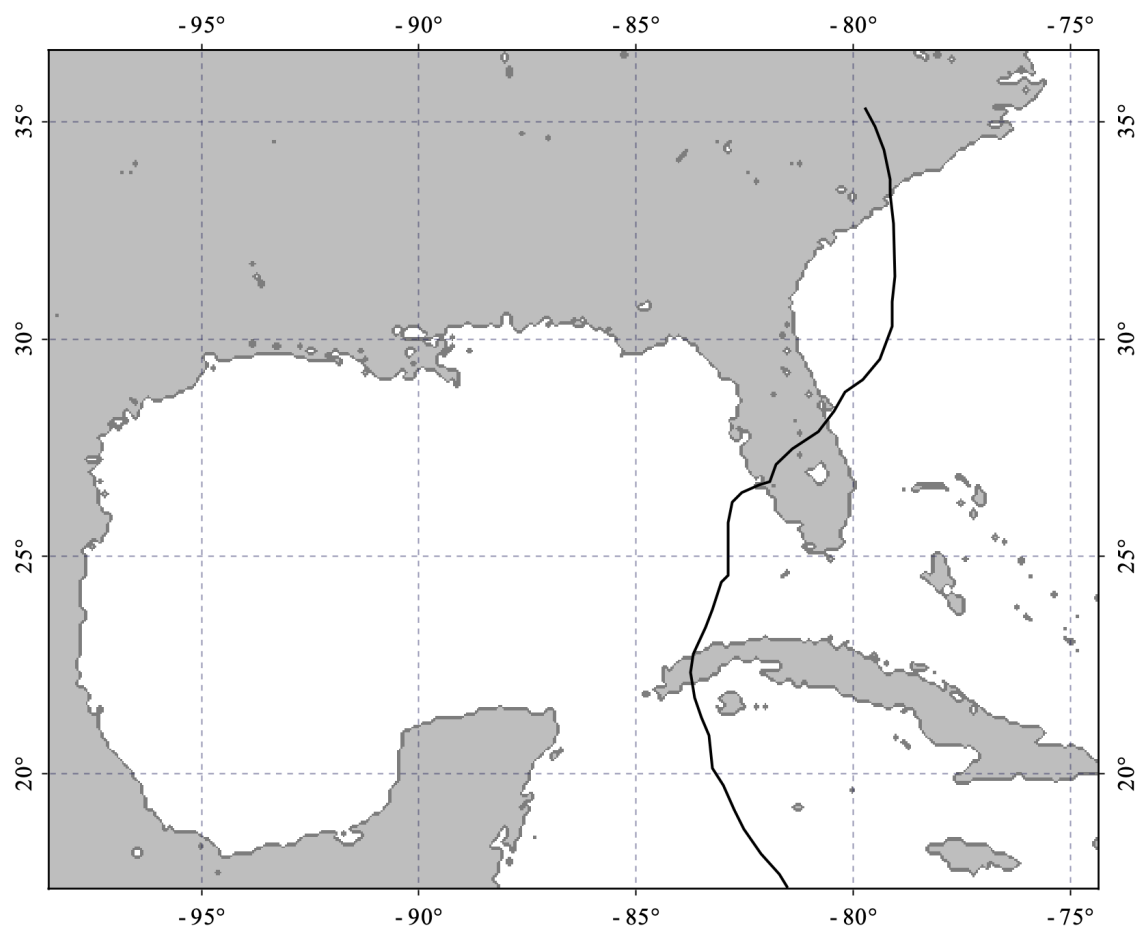
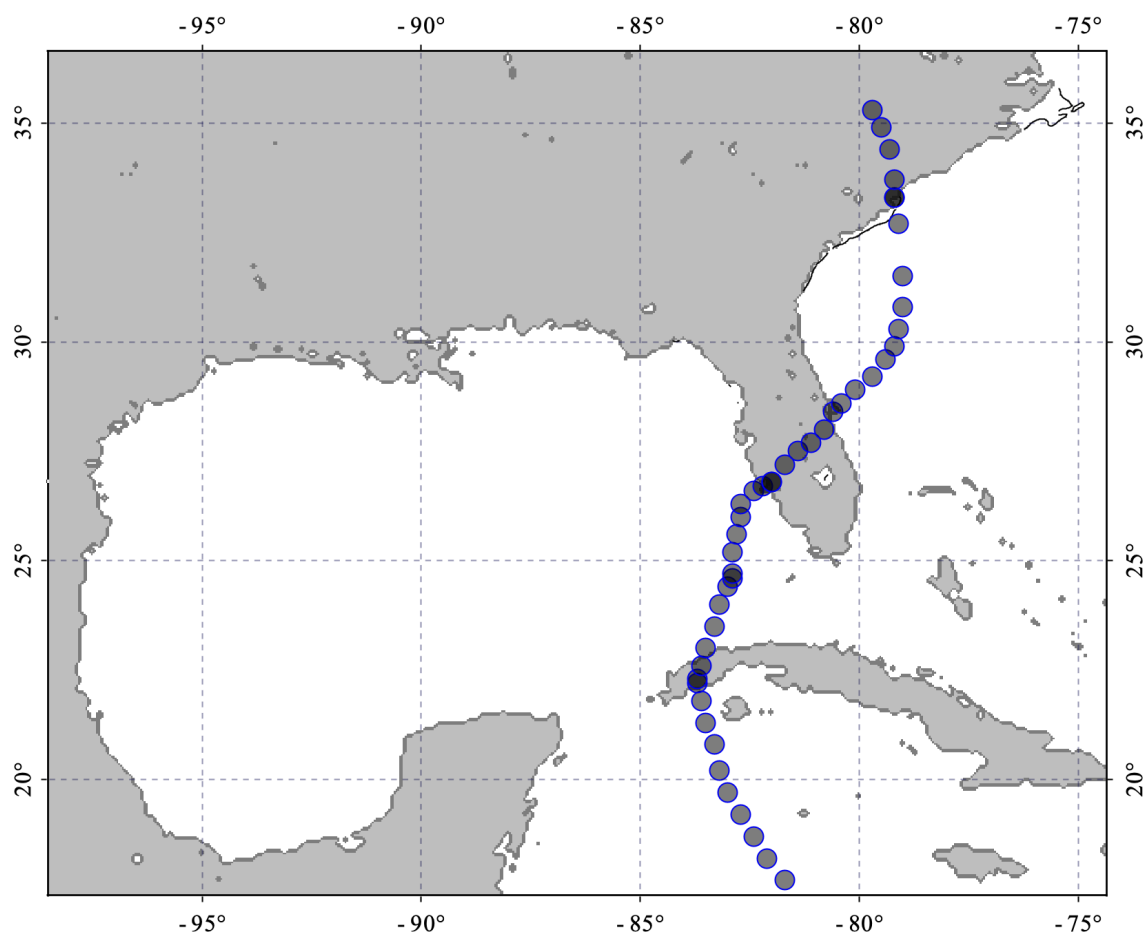
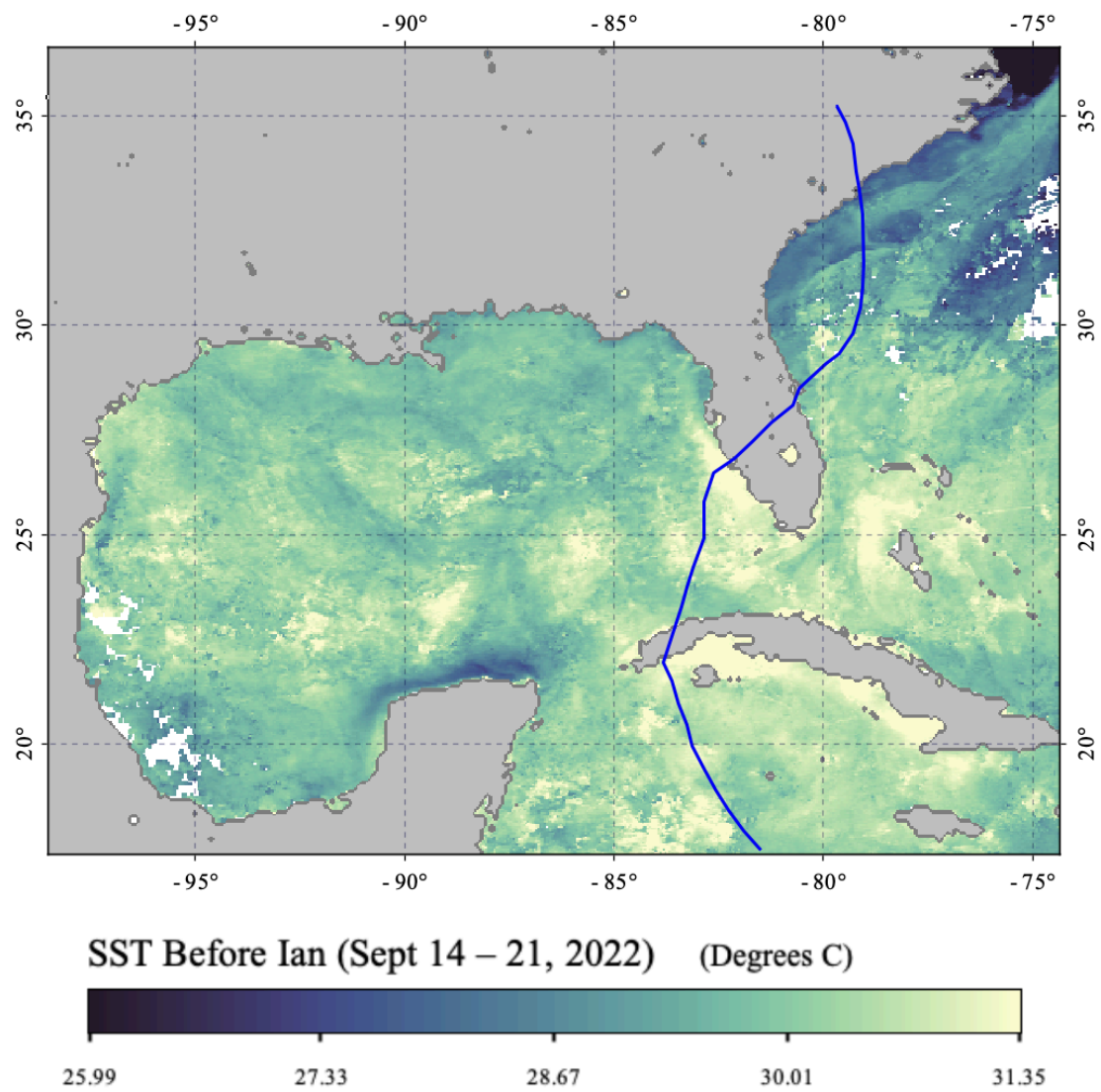
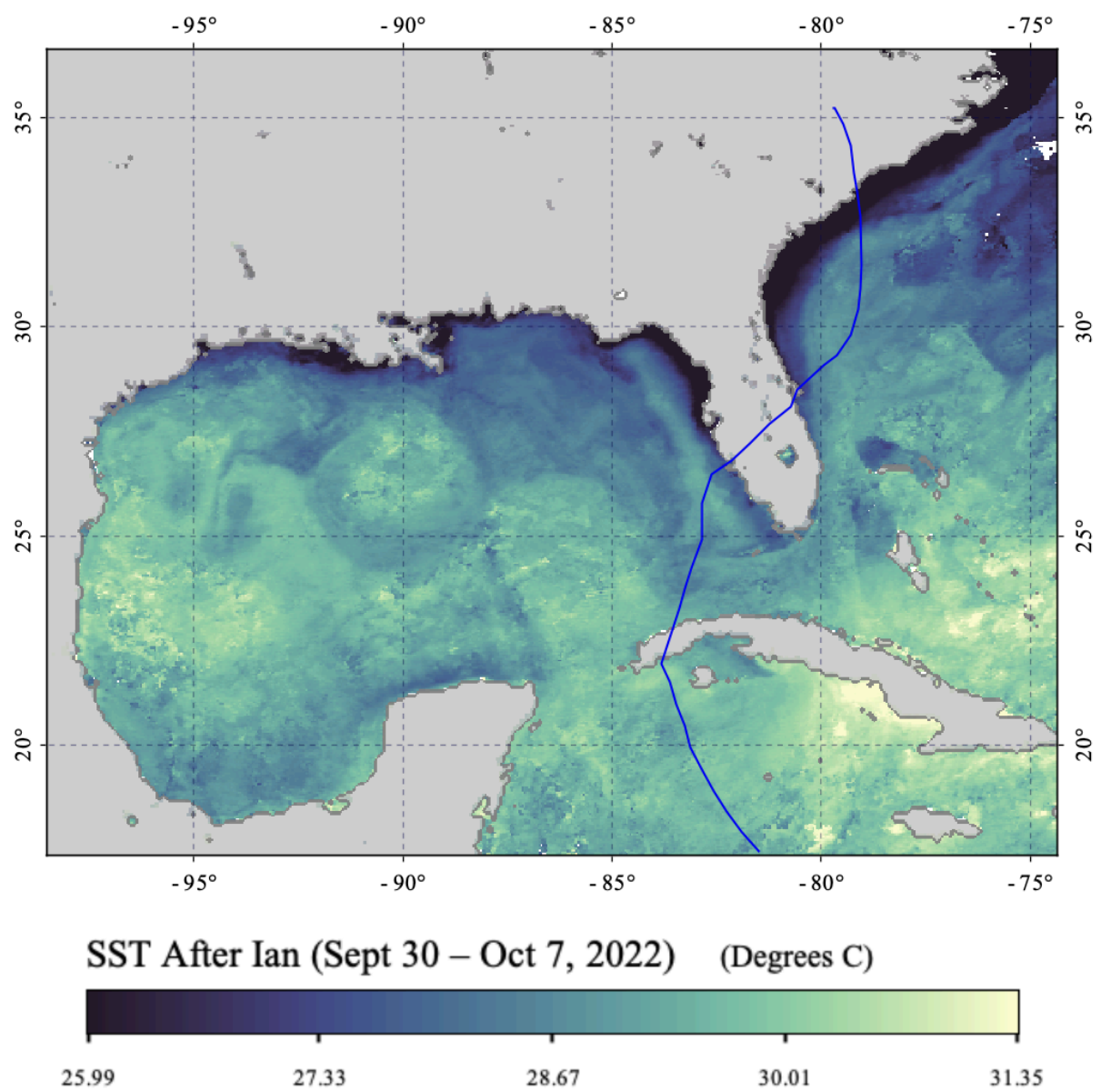


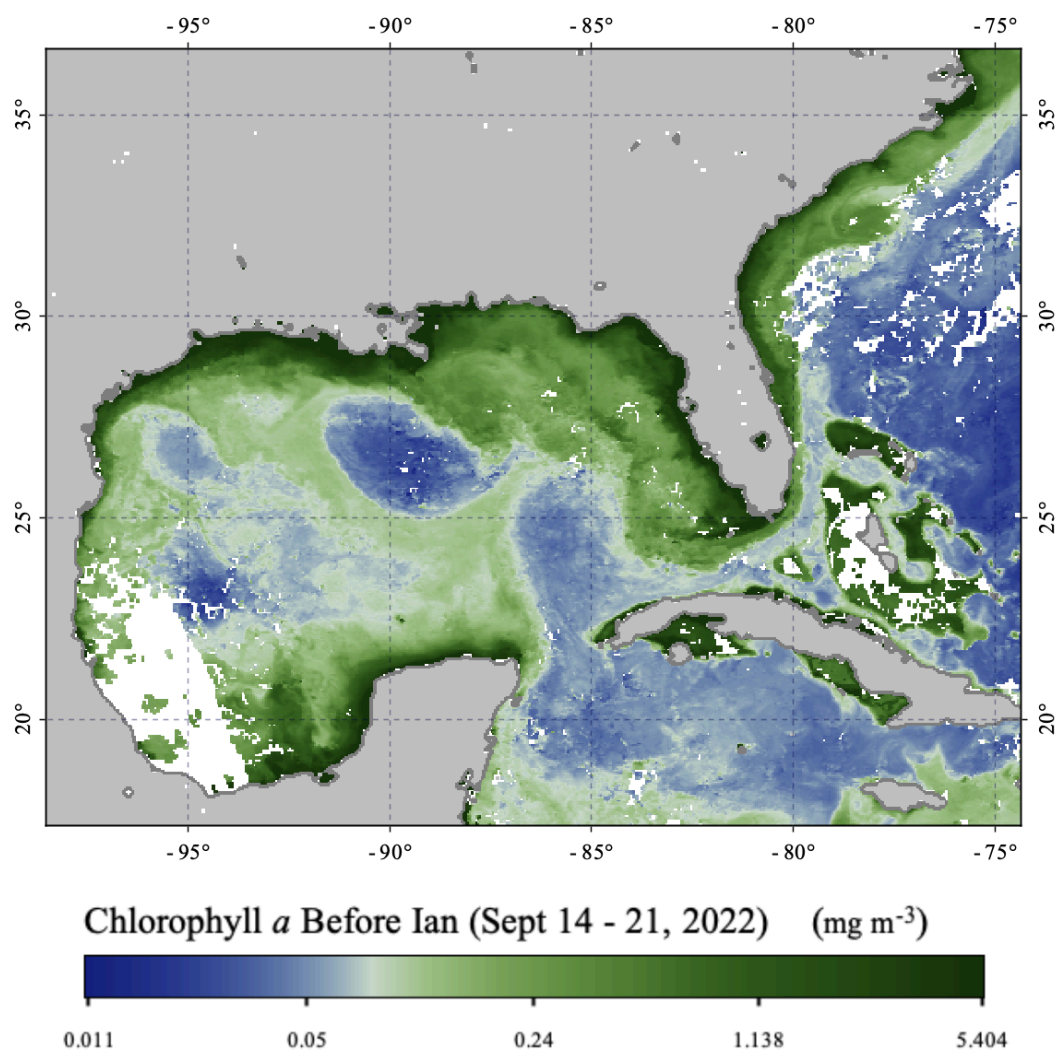
Figure 3: Mapped Hurricane Ian Points (SEADAS)

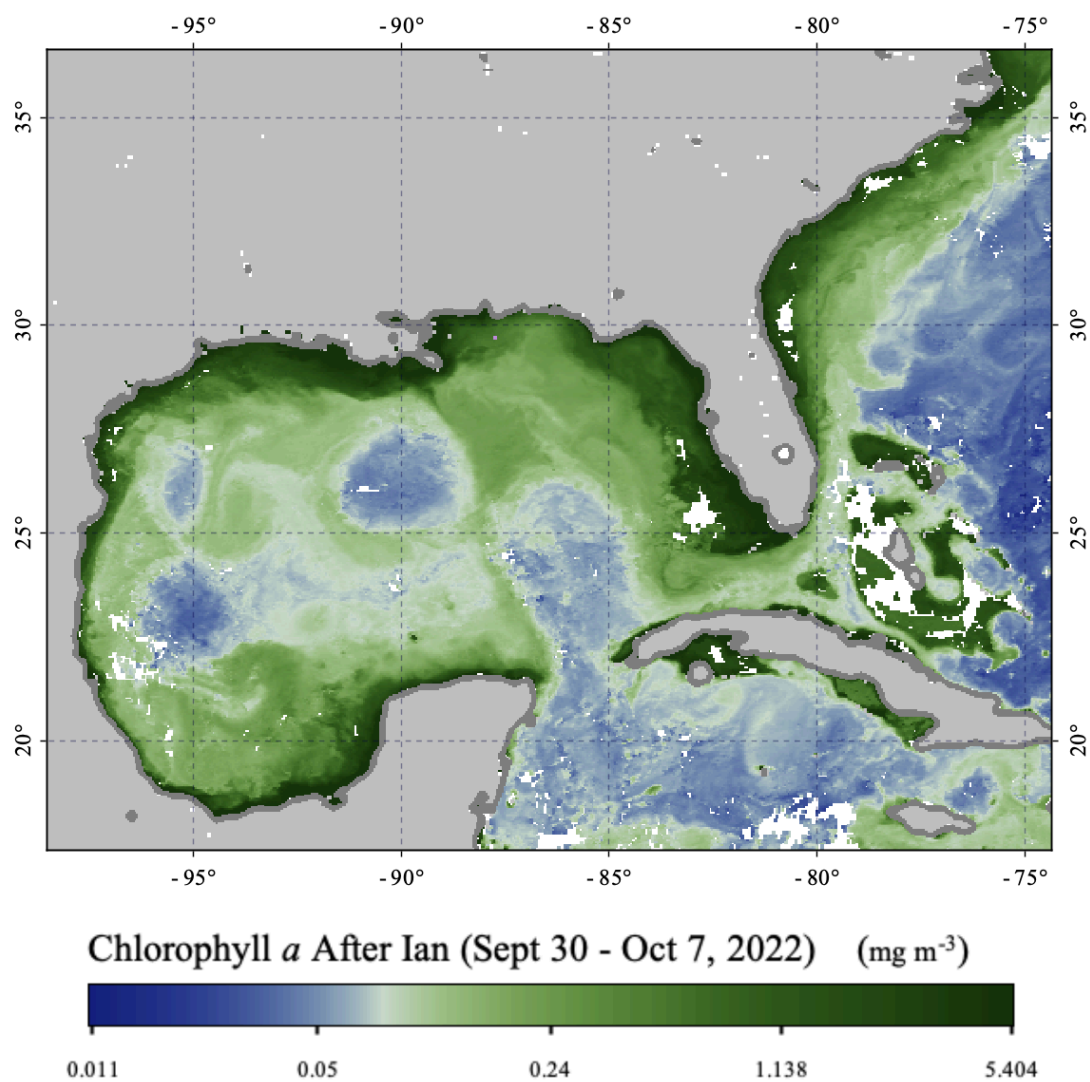


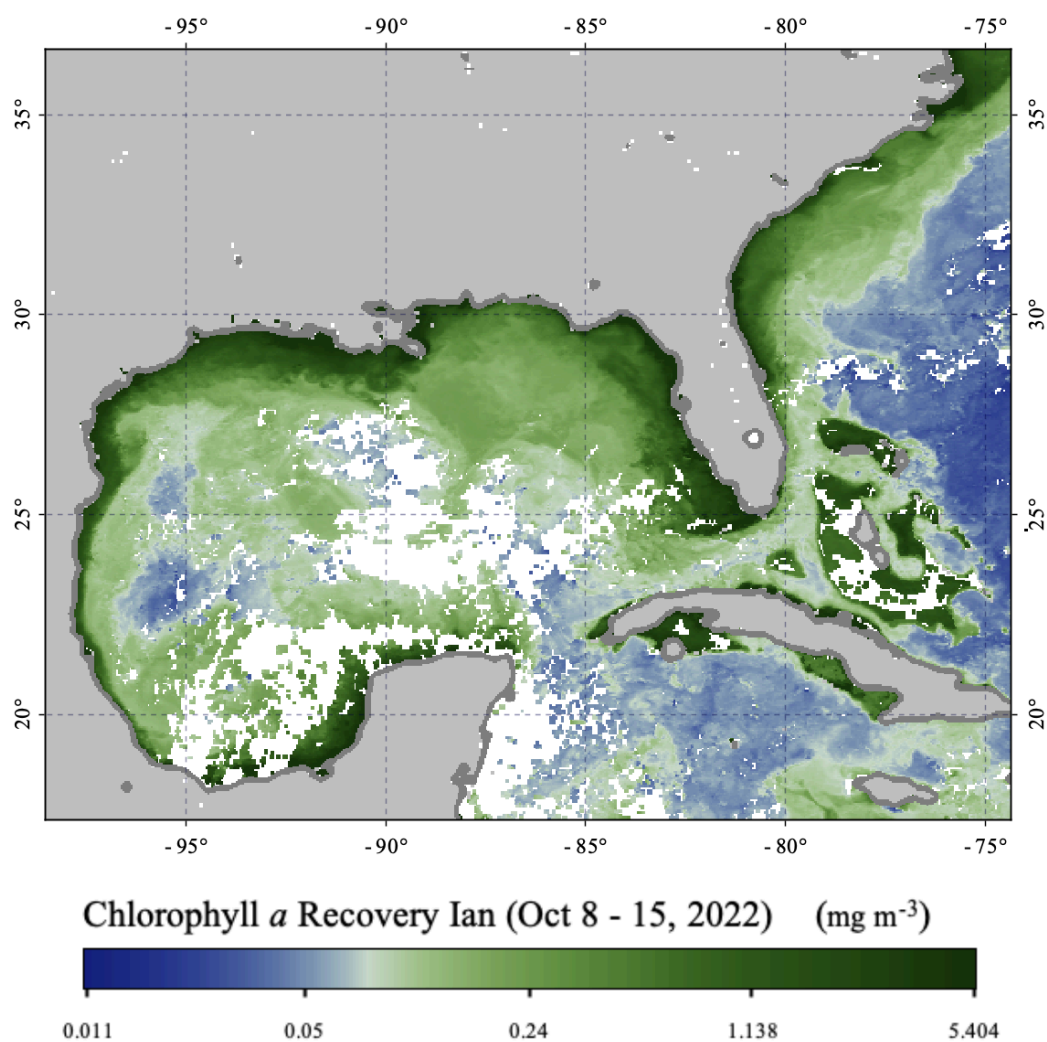
Figures: 5-9: SST and Chlor across 8 Day Intervals





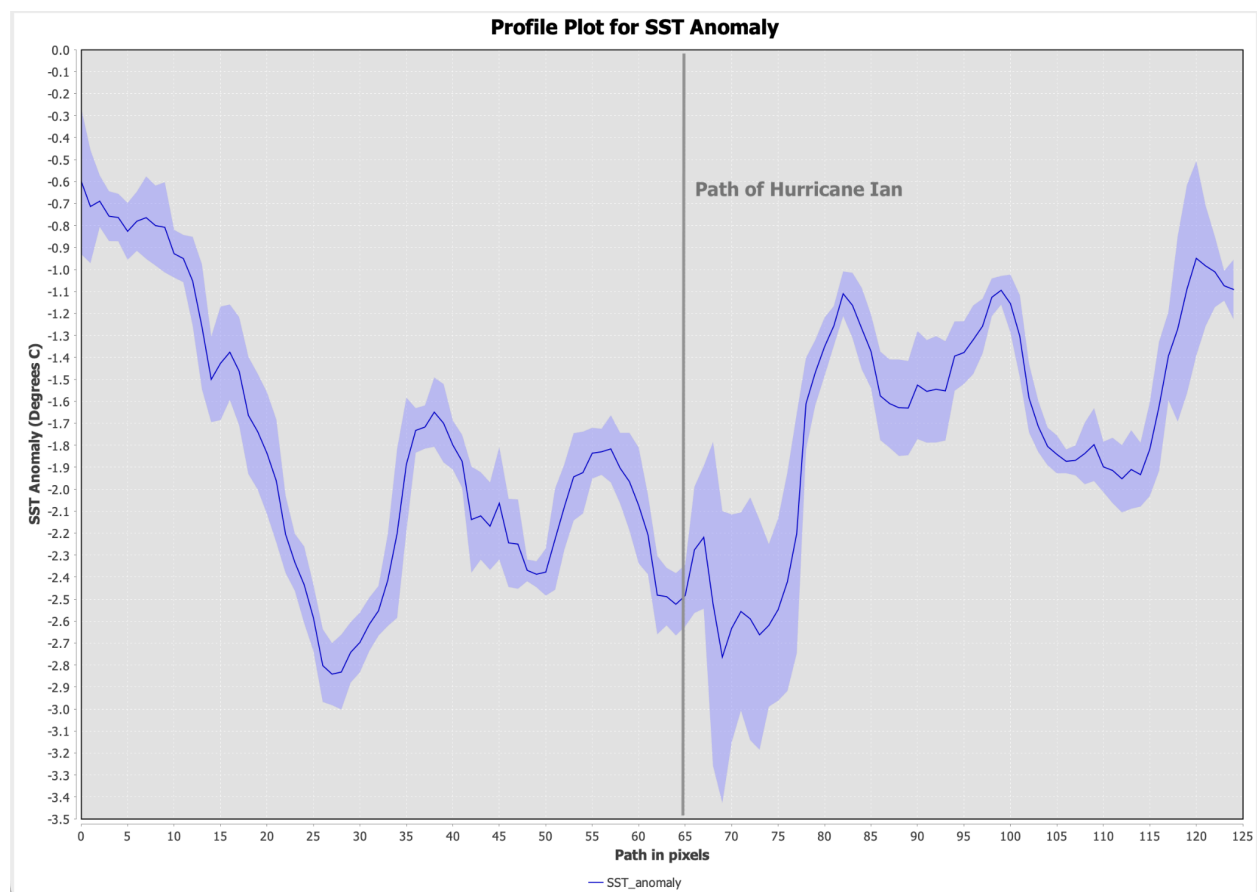


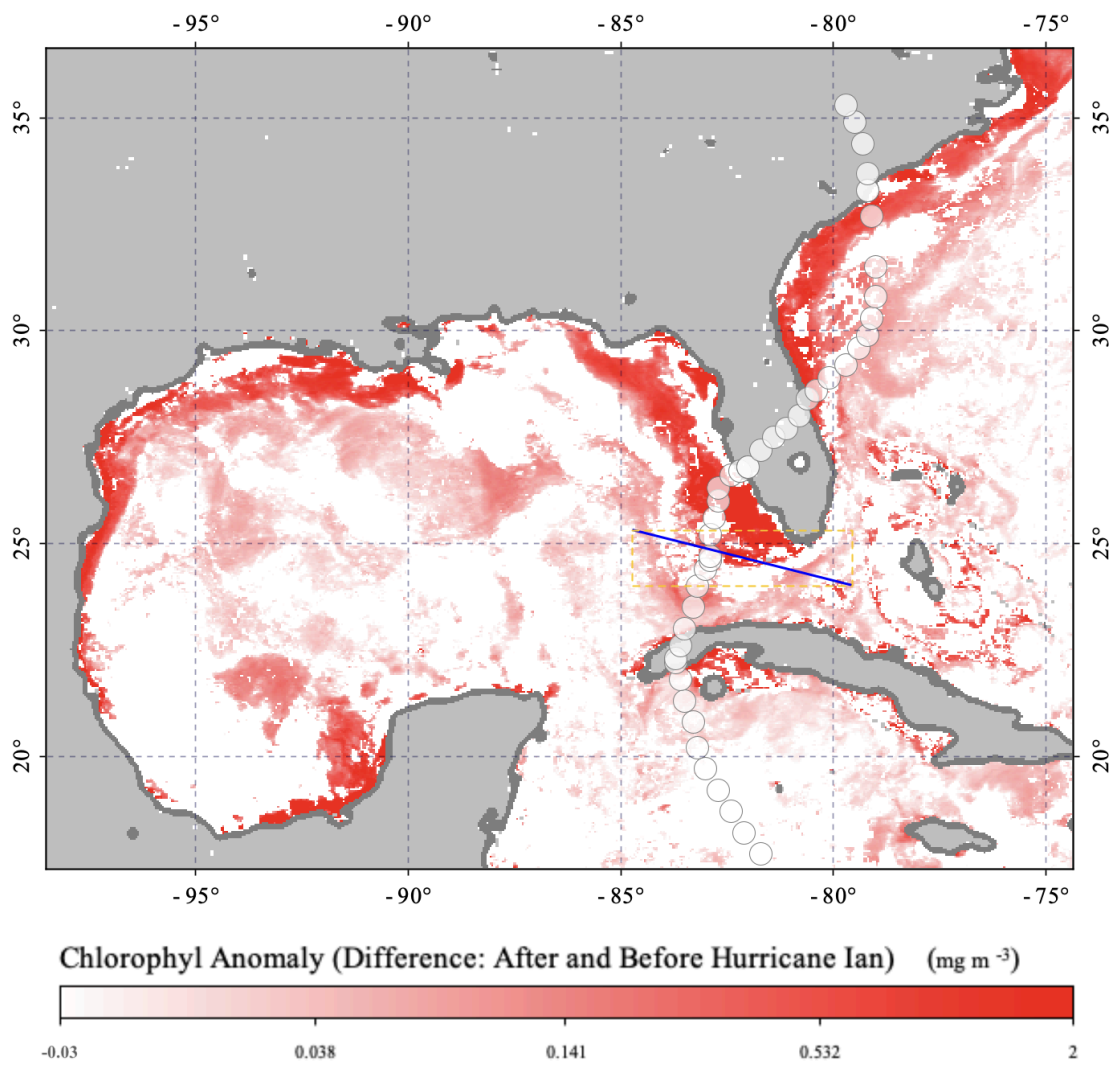


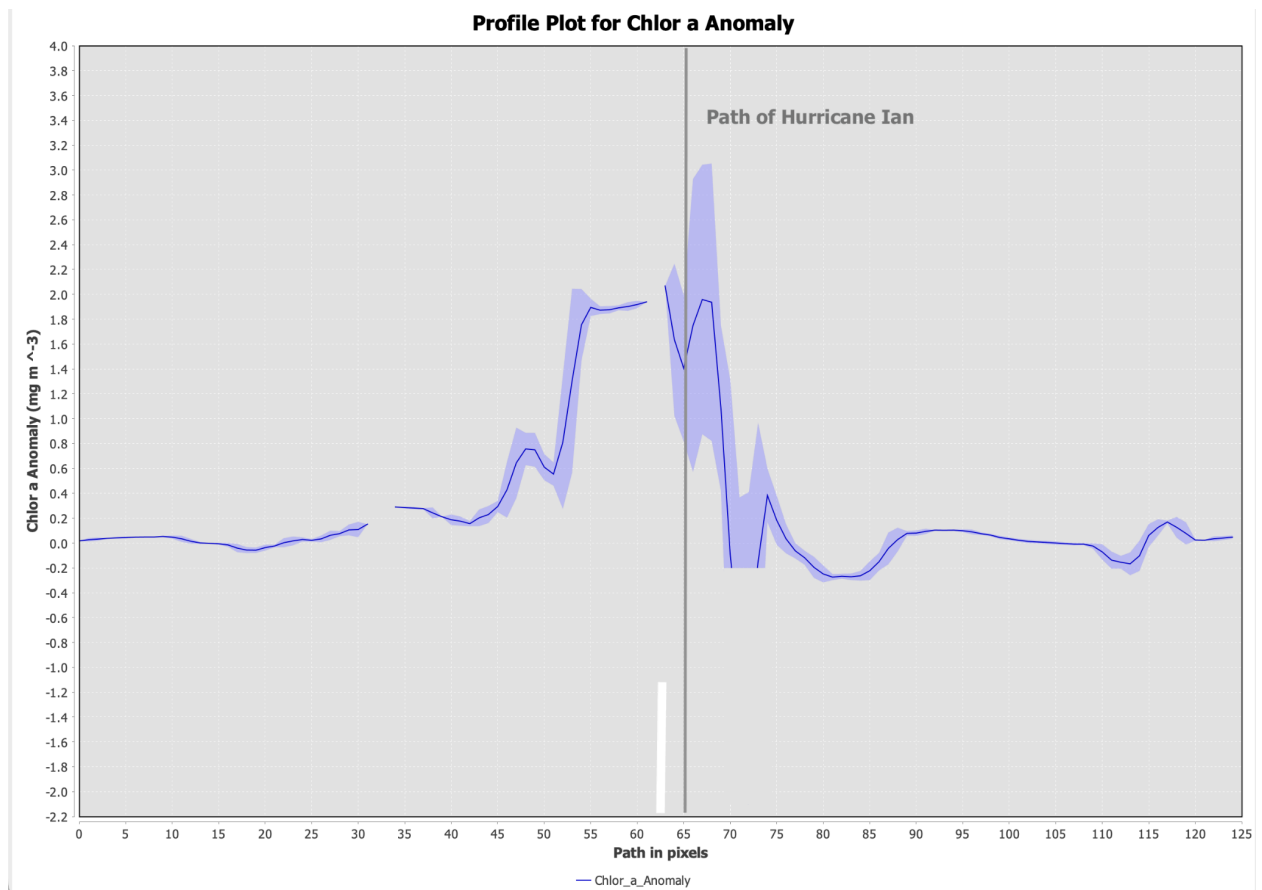


Figures: 11 - 14: SST and Chlor Anomaly Maps and Profile Plots



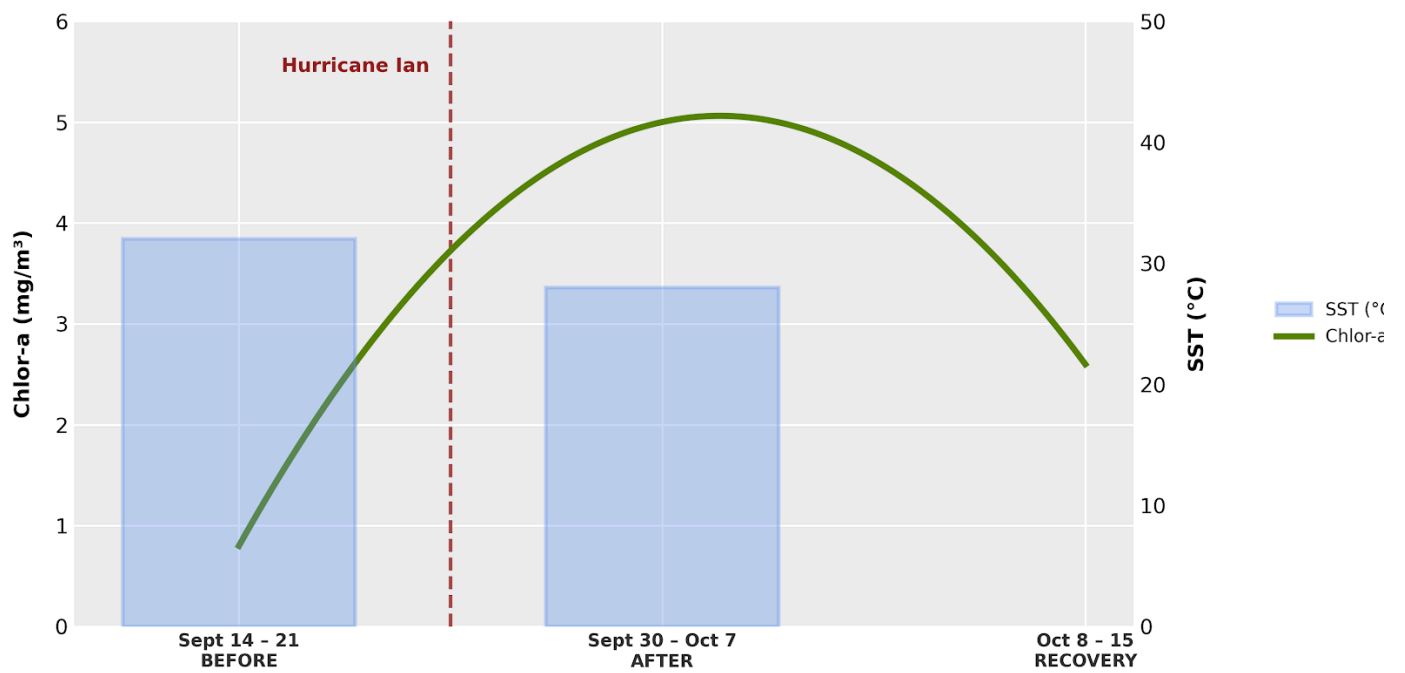






Figures: 15 - 16: Time Series Lag Graphs

SST and Chlorophyll-a Time-Series: West Florida Shelf



SST and Chlorophyll-a Time-Series: East Florida Shelf

